Week 8 web page: delivery guide

Instructor notes for Week8_PosteriorsAndOptimization.html

General

- Page: https://kevintshoemaker.github.io/NRES-746/Week8_PosteriorsAndOptimization.html. Open it before class; the widgets need no internet once loaded.
- For live demos, run the matching section of `CODE_Week8_PosteriorsAndOptimization.R` (same seeds, same numbers). The optimization part reuses `titer` from the first part's bonus section, so run the script from the top (or at least the `library(emdbook)` and `titer` lines).
- Zoom the browser to 125-150% for the projector.
- Questions on the page have no answers. Answers are below; *Land:* is where discussion should end up.
- Readings for Canvas: Bolker 4.3.3 (printed pp. 149-151), 6.2.2 (pp. 237-243), 6.4.2 (pp. 258-260), 7.1-7.2.2 (pp. 294-302).
- WODs: *How Much Did the Prior Matter?* (in class, Monday); *Newton's Method by Hand* (take-home, Monday to Wednesday); *Curvature by Hand* (optional, not collected, handed out Wednesday). All have keys.
- No lab this week (final project proposal meetings).

As taught (Oct. 12-14, 2026)

- **Monday** (from the post-lecture announcement, added Nov. 4): the WOD (students sketched the likelihood behind each prior/posterior pair), then the germination example (`#germination`), and summarizing a posterior (`#summaries`). Now on the page from the announcement: the "uncomfortable" point that every posterior, even a flat-prior one, assumes a prior belief on some scale (end of `#wod-priors`); flat vs. weakly informative/vague/diffuse vs. regularizing vocabulary, which the instructor forgot to say in class (`#stuck`); why the mean suits a right-skewed posterior, and HPD (most logically consistent) vs. quantile (most common) intervals (`#summaries`). Housekeeping: proposals due Monday, proposal meetings in the lab slot, the Newton take-home due Wednesday, Bolker Ch. 7 for Wednesday except the MCMC sections. The announcement doesn't mention sections 5-6 or the discussion.
- **Wednesday:** opened with why Bayesian methods are popular (section 6 of the first part), then the take-home debrief, the workflow, and Newton's method. Per the post-lecture announcement (added Nov. 4): finite differences (`#finite-differences`) were **not** reached and probably won't be revisited (the take-home WOD covered them; offer

to go over them if asked). The instructor bungled the board derivation of the Newton step from the linear approximation of the first derivative; the page's `#newton` now writes it out ($\text{NLL}'(\theta_k + \Delta) \approx \text{NLL}'(\theta_k) + \text{NLL}''(\theta_k)\Delta = 0$), with the gradient/curvature ingredients and the one-step-for-a-parabola point added to `#opt-big-picture` and `#newton`. The announcement's list of Bayesian selling points (messy data, more reliable convergence, missing data and latent variables, no quadratic approximation with MCMC) was merged into `#why-bayes`. *Curvature by Hand* went out as optional. Nelder-Mead and simulated annealing were not reached; they opened Week 9 and are on the Week 9 page. BFGS presumably wasn't reached either.

- The page was built after the week (Oct. 28) from the PDF NOTES (now in `lecture_notes/archive/`) and the legacy LECTURE5 page (now in `archive/`). New on the page: the two widgets, the grid-posterior and two-peak bonuses, the "shape of the valley" bonus, and the optimization discussion prompts.

Part 1: Priors and posteriors (Monday, ~35 min)

WOD (~8 min, before the lecture). Tortoise analysts, same 21 of 30 data, three priors. Then show section 2's figure (likelihood added).

- "Analyst B's prior comes from a predator-proof pen. Is it a reasonable prior for juveniles in the wild? What would you report to a manager who needs one survival rate?" *Land:* no; a pen removes predation, a major cause of juvenile death. Report A (mean 0.68, interval 0.51 to 0.82), mention C (0.62) as a sensitivity check, and say 30 juveniles aren't enough to make the prior irrelevant. Let students spot the pen problem; if no one does, ask where each prior came from.
- Trap to name: B has the narrowest posterior, but that's the prior.

Big picture (~3 min)

- Point at FIT. Same model and likelihood as maximum likelihood; the answer becomes a distribution.
- Board: Beta prior times binomial likelihood, add the exponents. Add counts.

Germination (~10 min)

- Run the chunk: flat mean 0.738 (mode 0.75 = MLE); supplier 0.624.
- Weighted average: prior worth 200, data 40, so 0.625.
- Widget stops, in order (predict before each):
  - Supplier + 30 of 40 (default): mean 0.624, 0.56 to 0.68; data carry 17% of the weight.
  - Flat + 30 of 40: 0.738, 0.60 to 0.86; posterior = likelihood.
  - Supplier + 300 of 400: 0.699; data carry 66%.
- "Starting from the supplier's prior and 30 of 40, how many seeds (at 75%) to land halfway between 0.60 and 0.75?" *Land:* about 200, the prior's worth; set seeds tested to 202 (mean 0.676).

Summaries (~4 min)

- Flat: mean 0.738, median 0.742, mode 0.750; central 0.597 to 0.858; HPD 0.605 to 0.864.
- Read the interpretation sentence aloud. This is the one people want to say about a confidence interval.

Where new Bayesians get stuck (~8 min)

- Core: likelihood $\neq$ posterior (area 1/41); credible vs. confidence (pikas 1.82 to 4.61 vs. 1.91 to 4.74). Flat-on-which-scale is a quick callback (covered in Week 7). Others if time.
- If asked what prior gives the pika credible interval: Gamma(1, 0), flat on λ and improper; flat on $\log \lambda$ gives 1.78 to 4.54.

Why so popular (~5 min). Read the bullets; run the odds chunk (1.48 to 6.04; P(p > 0.6) = 0.97). Hand out the Newton take-home.

Discussion

- "Supplier mean 0.62, flat 0.74. Manager: 'keep the supplier's numbers out.' Reasonable? What would you report?" *Land:* partly. Earlier lots are real information, and 40 seeds is little, but lots differ (storage age, source, test conditions) and the prior outweighs the data five to one. Report both (a sensitivity analysis), perhaps a down-weighted prior in between (worth 20 seeds: Beta(13, 9), mean 0.69). Whether past lots resemble this one is ecological. Note the supplier's numbers are *lower* than the crew's, so self-interest isn't the issue. A hierarchical model of lot-to-lot variation is the principled fix (later).
- "How would a Bayesian get an interval for $-a/b$? Compare with a profile interval." *Land:* compute $-a/b$ for every posterior draw and take quantiles; no reparameterizing or constrained optimization. With many parameters this is much easier.
- "LR interval for the pikas is 1.82 to 4.61. 'So a 95% chance the true density is in that range?' " *Land:* not for a confidence interval (it describes the procedure). With a flat prior a Bayesian can say 95% probability it's between 1.91 and 4.74, given the model and prior.

Bonus sections. Grid posterior (90,000 evaluations; leads to MCMC). Two-peaked posteriors: the HPD region splits; the mean lands in the valley.

Part 2: Derivative-based optimization (Wednesday, ~40 min)

Take-home debrief (~8 min). Run as a discussion. Key in `wod/answer_keys/WOD_Week8_Lecture1_Newt`

- 3a: a student puts $\lambda^* = \lambda - \text{NLL}'/\text{NLL}''$ on the board (straight line through the current slope hits 0).
- 3b: 2, 2.667, 2.963, 3.000; error roughly squares each step.

- 3c: from 7 to -2.33; collect fixes (shorter steps, better start, bounds, log scale).
- 2a/2b: the forward slope at 2 (-2.03) matches the exact slope at 2.25 (-2); second difference 4.65 vs. 4.5.

Big picture (~5 min). First time the full workflow was drawn: walk the boxes once, then point at FIT. Grid cost: 14,400 evaluations here; 30^{10} for ten parameters (19,000 years at 1 ms).

Newton in one dimension (~8 min). Run the chunks. Widget stops:

- Start 2, exact, λ: Step $\times$ 3 gives 2.667, 2.963, 3.000. Point at the parabola (top) and the straight line (bottom).
- Start 7: one step to -2.33; the widget stops and says why.
- Switch to log λ (start 7): 3.953, 3.106, 3.002, 3.000.
- "Start at 7 on the λ scale, then on the log scale. Why can't the log-scale version overshoot past zero?" *Land:* every θ maps to a positive $\lambda = e^{\theta}$, so even a long step lands at a valid value; there's no edge to jump past.

Slopes without calculus (~5 min). Forward -2.03 vs. exact -3 at 2, exact -2 at 2.25 (the chord's midpoint). Widget, start 2, finite differences: $\Delta = 0.5$ gives slope -2.03, curvature 4.65, next 2.438 (vs. 2.667 exact); $\Delta = 0.05$ gives 2.642. Mention `gr`.

Newton in 2D and BFGS (~10 min). Board: gradient, Hessian, $\theta - H^{-1}g$. Good start converges in ~6 steps; bad start lurches (NLL goes up at step 7).

- "Nelder-Mead 37.667 (convergence 0); BFGS 37.670 (convergence 1), estimates look close. Is the BFGS fit done?" *Land:* no; 1 means it hit the iteration limit (100). Always check `$convergence`. Restart from where it stopped, use a better start or `parscale`, or fit on the log scale (33 evaluations, convergence 0).
- **Behind?** Skip the 2D Newton path table and go straight to BFGS.

Discussion (if time; otherwise Week 9's opening)

- "Two starts end at NLLs 34.2 and 36.8, both convergence 0. Which do you report?" *Land:* the lower, after checking it (refit from it, try another method). The gap means more than one minimum or a run that stopped on a flat stretch; try more starts and look at a slice between them.
- "Replace any negative guess with 0.001?" *Land:* from 0.001 the update roughly doubles λ each step (0.002, 0.004, ...), so it takes about 13 steps to get near 3. It recovers, slowly and by luck. Backtracking (halve the step), bounds, or the log scale are principled fixes.
- "Why not a step of 10^{-15}?" *Land:* rounding error. Computers keep about 16 significant digits, so $f(x+\Delta) - f(x)$ loses almost all of them when Δ is tiny. At $\lambda = 2$ the forward slope is -2.998 with $\Delta = 0.001$, -3.0000 with 10^{-8}, -3.0003 with 10^{-12}, and -1.78 with 10^{-15} (Bolker p. 300).

Bonus: the shape of the valley. Correlation -0.995; mean = shape $\times$ scale = 6.92 is what the data know. Fitting mean and shape removes the correlation. Bolker's 7.4 checklist.